mahi-gpu

8-thread SIMT GPGPU core for SkyWater sky130

Datasheet and Architecture Reference Manual

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Design source	repository commit `dadec7f`, directory `ASIC/src/rtl`
Top module	`gpu_asic_top`
Target process	SkyWater sky130A, `sky130_fd_sc_hd`

Status of this document

This manual describes the RTL *as implemented*. It deliberately states what is **not** yet true of the chip: the **mini profile** has a routed layout that passes the checks available in the repository, but the full-size profile has never been routed (Chapter 5), the SRAM macros are timed at the typical corner only and are black boxes in LVS, nothing has been simulated on the gate-level netlist of this design, and barriers require warp-uniform control flow (a `SYNC` while diverged faults). See Chapter 6 (Errata) before relying on any number in Chapter 5.

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Chapter 1

Overview

1.1 Executive summary

mahi-gpu is a compact general-purpose GPU core built around the SIMT execution model. One *core* runs **2 warps** of **4 lanes** each (8 hardware threads); every lane has its own integer ALU, single-precision FMA unit, 32-entry register file, program counter and return stack. Warps are interleaved by a round-robin scheduler on a classic five-stage in-order pipeline (IF, ID, EX, MEM, WB). A 5-stage pipelined IEEE-754-style FMA unit executes `FADD`, `FSUB`, `FMUL` and `FMA` under a per-warp register scoreboard. Memory is reached through a banked shared memory, an instruction cache, a 2-way write-through L1 data cache, and a 4-way write-back L2 cache with sector dirty bits and a victim write buffer. The host controls the device through a 32-bit Wishbone slave.

The ASIC build (`-DTARGET_ASIC`) is a size-reduced profile of a configurable design that also targets FPGA and simulation; all sizes in this manual are for the ASIC profile unless stated otherwise.

1.2 Key characteristics

Characteristic	Value
Process / library	SkyWater sky130A, `sky130_fd_sc_hd` (high density), 1.8 V nominal
Supply corners analysed	1.60 V/100 °C (ss), 1.80 V/25 °C (tt), 1.95 V/−40 °C (ff)
Flow	Yosys (via sv2v) → OpenROAD (floorplan, place, CTS, route) → Magic/KLayout, driven by LibreLane (§5.5)
Clock constraint (target)	25 MHz (40 ns period), single clock `clk`, synchronous reset `reset`
Cores / warps / lanes	1 core / 2 warps / 4 lanes = **8 threads**
Pipeline	5-stage in-order (IF, ID, EX, MEM, WB); FPU: 5-stage, fixed latency
Register file	32 × 32-bit per thread; `R0`–`R28` general, `R29`–`R31` read-only
Instruction word	32 bit, fixed width; 37 opcodes defined (0–36)
Program memory	4 K words (12-bit word address), external, 128-bit block interface
Global data memory	16 K words (14-bit word address), external, 128-bit block interface
Shared memory	**64 words** (256 B), 4 banks × 16 words
L1 I-cache / L1 D-cache	16 lines each; 16-byte lines (4 words); I: direct-mapped, D: 2-way, write-through
L2 cache	64 lines (16 sets × 4 ways), 16-byte lines, 4 sectors/line with dirty bits, PLRU, write-back; 4-entry victim write buffer
Host interface	32-bit Wishbone slave, single-cycle ack; **108 signal pins** (excl. power)
Synthesised logic (current RTL)	151 502 instances, 24 269 flip-flops; 1.80 mm^2 cells + 0.85 mm^2 for three SRAM macros (§5.5)
Die size (floorplan)	3.0 mm × 3.0 mm (9 mm^2) in the current config; 2.5 mm in the earlier runs

1.3 Conventions

Numbers: `0x` prefix hexadecimal. Bit ranges `[hi:lo]` are inclusive. Addresses in the memory system are *word* addresses unless stated. “Lane” and “thread” are used interchangeably within a warp. Source citations use the form `file.sv:line` relative to `ASIC/src/rtl`. The words *limitation* and *erratum* refer to the register in Chapter 6 (IDs **L1**–**L14**).

Chapter 2

System architecture and memory map

2.1 Block diagram

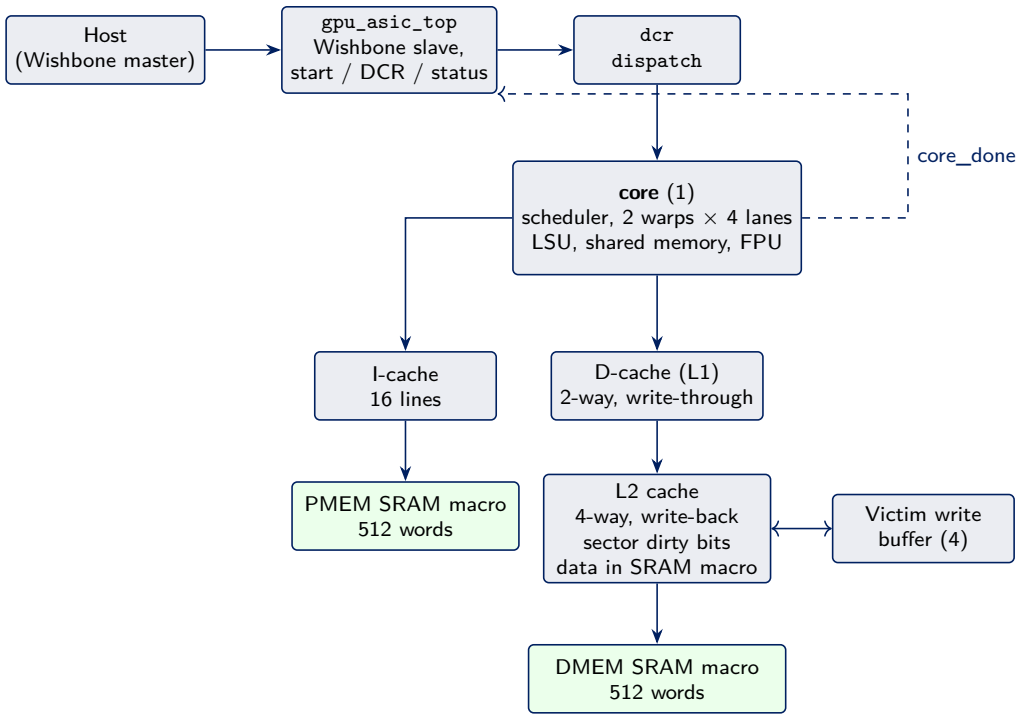


Figure 2.1: System block diagram of the ASIC profile. Green boxes are on-chip OpenRAM SRAM macros (a third, inside the L2 cache, holds its data array); the host reaches PMEM and DMEM through the Wishbone window.

The I-cache has its own path to the program-memory controller and does *not* pass through the L2. Both memory interfaces use a native 128-bit block protocol (valid/ready/address/data/strobe) through `controller.sv`. On chip, `gpu_mem_subsys` terminates them on two 512×32 SRAM macros (one block = four sequential word accesses). An AXI4 adaptor exists in the repository but is **not** instantiated in the ASIC top level.

2.2 Host interface: Wishbone slave

`gpu_asic_top` is a 32-bit Wishbone classic slave with a registered, single-cycle acknowledge (`wb_ack_o` rises one clock after a request with both `wb_cyc_i` and `wb_stb_i` set).

2.2.1 Register map

The base address is `0x3000_0000`. `wb_adr_i[31:12]` selects the page: `0x30000` is the control/status page, `0x30001` the memory window; every other address is acknowledged, reads as 0 and ignores writes. Every access is acknowledged (a window access that is refused still returns an ack). While the GPU runs, window accesses are refused: reads return `0xBAD0_BAD0`, writes are dropped and STATUS[3] is set (erratum L16).

2.2.2 Internal Device Control Register (DCR) block

The DCR (`dcr.sv`) has an 8-bit address / 8-bit data write port. **Only address `0x00` is reachable from the chip pins**: the Wishbone wrapper always drives address `0x00` (erratum L13). The remaining registers exist for integration and simulation.

Signal	Dir	Width	Function
clk, reset	in	1, 1	clock, synchronous reset
wb_cyc_i, wb_stb_i, wb_we_i	in	1 each	cycle, strobe, write enable
wb_sel_i	in	4	byte select (memory-window writes only)
wb_adr_i	in	32	address (bits [31:12] select the page, [7:0] the register)
wb_dat_i	in	32	write data
wb_dat_o	out	32	read data (registered)
wb_ack_o	out	1	acknowledge
gpu_done_pin	out	1	mirrors the done status bit
gpu_fault_pin	out	1	mirrors STATUS[4] (a warp faulted; sticky until cleared), intended for an in

Table 2.1: Pins of gpu_asic_top: 108 signal pins in total (73 inputs, 35 outputs), excluding power.

Address	Access	Name	Description
0x3000_0000	W	CTRL	[0] START (level; stays set until written 0); [1] SOFT_RESET (8 cycles; clears DCR, scheduler and all caches, keeps SRAM contents; drops START; other bits ignored); [15:8] THREAD_COUNT (written to internal DCR 0x00)
0x3000_0000	R	CTRL	{16'b0, THREAD_COUNT[7:0], 7'b0, START}
0x3000_0004	R/W1C	STATUS	[0] DONE; [1] BUSY; [2] GPU_OOB (sticky: data access outside the 512 words); [3] HOST_BLOCKED (sticky); [4] FAULT (sticky: a warp faulted); write 1 to clear [2]/[3]/[4]
0x3000_0008	R	FAULT_INFO	{PC[15:0], WARP[7:0], 4'b0, CAUSE[3:0]} of the first fault since STATUS[4] was last clear; CAUSE 1 = division by zero, 2 = bad address, 3 = SYNC while diverged
0x3000_1000–17FF	R/W	PMEM	512 words = GPU instruction words 0..511, byte strobes honoured
0x3000_1800–1FFF	R/W	DMEM	512 words = GPU data words 0..511, byte strobes honoured

Table 2.2: Wishbone register map as implemented (gpu_asic_top.sv:27-38).

2.3 Address spaces

2.4 Reset and launch sequence

1. Assert reset, then release.
2. Host writes CTRL with START=1 and THREAD_COUNT in bits [15:8]. START and the DCR write occur in the same bus cycle.
3. The dispatcher computes ⌈THREAD_COUNT/8⌉ blocks. With one core, **blocks run sequentially**: each block resets the core and launches 8 threads (the last block may be partial); R29 holds the block id.
4. When all blocks have finished, the dispatcher raises flush_caches; the D-cache invalidates and the L2 writes all dirty sectors back through the victim buffer. DONE is set only after every flush completes, so external memory is coherent when DONE reads 1.
5. DONE clears when the host writes START=0 and the dispatcher is idle. By code inspection (not simulated), holding START=1 after DONE re-triggers the flush loop, so software should drop START after observing DONE.

DCR addr	Name	Function
0x00	THREAD_COUNT	total threads in the launch (8-bit, reset 0)
0x01	PMU_RESET	one-cycle pulse: clear all PMU counters
0x02	PMU_SNAPSHOT	one-cycle pulse: latch counters into snapshot registers
0x10–0x13	PMU_CFG0–3	5-bit event selector for counter 0–3 (see Appendix A)

Space	Size / width	Access	Notes
Program memory	4 K words, 12-bit PC	fetch only	word-addressed; I-cache fetches 4-word blocks; only words 0..511 exist on chip (L15)
Global data	16 K words, 14-bit address	LDR STR ATOM_*	only words 0..511 exist on chip (L15); the core faults accesses above 0xFFFF, but the bus carries only 14 bits, so 0x4000–0xFFFF **silently alias** (L14)
Shared memory	64 words	LDSH STSH	per-core, shared by all 8 threads; bank = addr[1:0]; fault if address $\geq$ 64
Host MMIO	2 registers + 4 KB window	Wishbone	§2.2

Table 2.3: Address spaces of the ASIC profile.

Chapter 3

Core microarchitecture

3.1 Pipeline

Stage	Module(s)	Function
IF	`fetcher`, `icache`	3-state FSM (IDLE, FETCHING, DONE) fetching one instruction at a time for the warp chosen by the scheduler. The held instruction is tagged with the PC it was fetched for (`fetched_pc`); it is accepted only if that PC equals the PC of the warp being issued, otherwise the fetcher drops it and refetches.
ID	`decoder`, ID latch	Combinational decode; the `id_*` latch registers instruction, PC, warp id and active mask. Register-file read with write-through bypass. A flush of the previous occupant's warp squashes the latch only if no fresh instruction is loading in the same cycle.
EX	`alu`, `fp32_fma`, `pc` (per lane)	Per-lane integer ALU and FMA, forwarding network (MEM, WB and LSU-return sources), NZP update, next-PC computation, branch/divergence/exception detection.
MEM	`lsu`, `shared_mem`	Memory instructions engage the LSU; the issuing warp parks in `WAITING_MEM` until its lanes complete. Other instructions pass through.
WB	`registers`	Three writers with fixed priority: FP writeback > LSU return > integer ALU.

Table 3.1: Pipeline stages.

3.2 Datapath

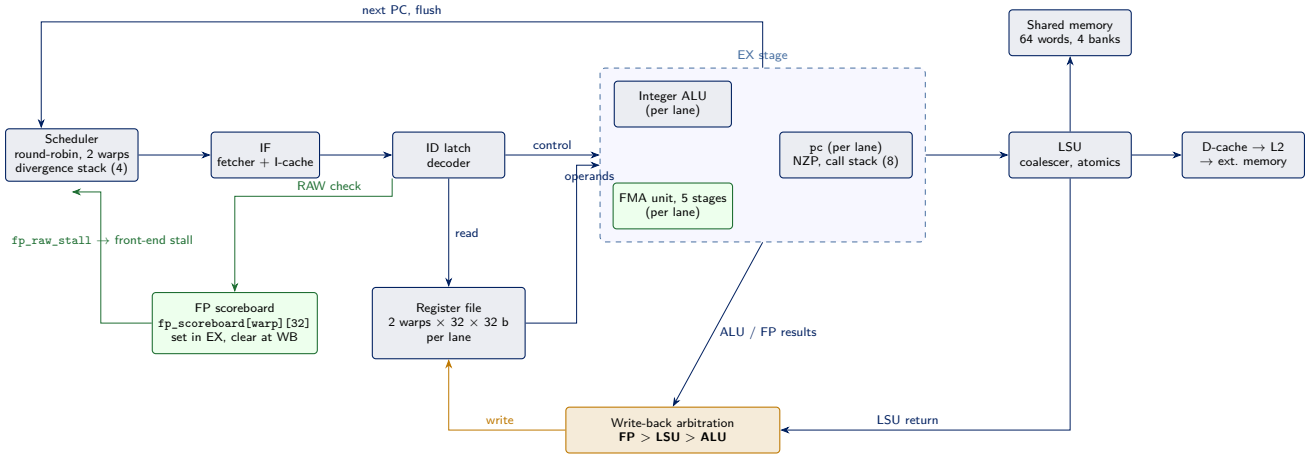


Figure 3.1: Core datapath. Blue: instruction/operand flow; green: FP hazard control; amber: write-back arbitration into the register file.

3.3 Warp scheduler

The scheduler (`scheduler.sv`) selects, every cycle and combinationally, the next `READY` warp with a non-empty active mask, round-robin from `rr_ptr`, excluding warps that are being flushed in the same cycle. Warp

states: `IDLE`, `READY`, `WAITING_MEM`, `WAITING_BARRIER`, `WAITING_FP`, `DONE_STATE`, `FAULTED`. Branch, call, divergence and exception outcomes are resolved in EX and redirect the front end through a one-cycle per-warp pulse `flush_warp_mask`; the squashed instructions are those between fetch and EX for that warp.

3.3.1 Divergence handling (SIMT stack)

Each warp has a 4-entry stack of `(PC, active_mask)` (`scheduler.sv:59-67`). In EX, the next PC of every active lane is computed. If all lanes agree, the warp continues. If they disagree, the lanes are split into two groups: group A (the group containing the lowest active lane) continues, and group B's `(target, mask)` is pushed.

- **Reconvergence** happens when a non-divergent branch of the running group targets *exactly* the PC on top of the stack: the entry is popped and the masks are merged.
- At `EXIT`, a non-empty stack is popped and the saved group resumes at its saved PC.
- There is **no** reconvergence at a join point or post-dominator. In practice lane groups run the remainder of the program serially until `EXIT`. Results are correct; a `SYNC` reached while the stack is non-empty faults (§4.5, erratum L1).

Maximum nesting is lane-count minus one (3 for four lanes), which fits the 4-entry stack. A profile with more than five lanes could overflow it (L7).

3.3.2 Call/return stack

Return addresses are held by each lane's `pc` instance: 8 entries per warp with a 3-bit pointer (`pc.sv:29-33`). `CALL` pushes `PC+1` and jumps to the absolute target; `RET_FN` pops. There is **no overflow or underflow detection**: the ninth nested call wraps the pointer and overwrites the oldest return address (erratum L6, demonstrated by `make limits`).

3.4 Floating-point unit

Each lane has a private `fp32_fma` with a fixed **5-cycle** latency from the EX stage to the register write. Pipeline

Instruction	Internal operation $(a \cdot b) + c$	Result
`FMA`	$a \cdot b + c$ with a=`rs`, b=`rt`, c=`rd`	`rs` × `rt` + `rd`
`FADD`	$a \cdot 1.0 + b$	`rs` + `rt`
`FMUL`	$a \cdot b + 0$	`rs` × `rt`
`FSUB`	$a \cdot 1.0 + (-b)$	`rs` − `rt`

stages: (1) mantissa multiply and exponent add, (2a) alignment, (2b) add/subtract with sign selection, (3) normalisation (leading-one detect), (4) pack. Numerics (this is *not* full IEEE-754; erratum L8):

- results are **truncated** (no rounding); denormal inputs read as zero and results below the normal range flush to ± 0;
- exponent 255 is treated as an ordinary exponent (no NaN/Inf propagation); overflow produces an infinity *encoding*;
- the product exponent is computed even when an operand is zero, so $0 \times 2^{127} + c$ drops low mantissa bits of c;
- exact cancellation yields a zero carrying the sign of the product term $(a \cdot b)$.

3.4.1 Scoreboard and hazards

`fp_scoreboard[warp][reg]` is a 32-bit-per-warp vector. An FP instruction sets its destination bit when it fires in EX; the bit clears when the result reaches write-back (fifth cycle). The ID stage compares the instruction's `rs`, `rt` and (when read) `rd` with the scoreboard; a hit raises `fp_raw_stall`, which stalls the front end and squashes the EX latch for that cycle. In addition, when an FP instruction reaches EX the *warp* is parked in `WAITING_FP` and its younger instructions are flushed (`scheduler.sv:200-204`); it resumes when its FP result is written back. Other warps keep issuing.

3.4.2 Register-file write arbitration

Priority 1 FP write-back, priority 2 LSU return, priority 3 integer ALU (`registers.sv`); lower priorities are suppressed when the same warp *and* register collide. Writes to `R29`–`R31` are ignored by all writers.

Why the ALU-versus-FP collision guard cannot be exercised

Removing the ALU-side FP-collision guard in a mutation experiment was *not* detected by 100 random programs. The design reason is that an FP operation parks its own warp in `WAITING_FP` and flushes its younger instructions, while older instructions have left the pipeline before the five-cycle FP result returns, so an ALU write to the same warp/register cannot coincide with the FP write-back. This is a design argument supported by the 0/100 mutation survival; it has **not** been formally proven.

3.5 Memory subsystem

3.5.1 Load/store unit and coalescing

The LSU (`lsu.sv`) accepts one memory instruction per warp and works through the active lanes with a state machine (`IDLE, COALESCE_READ, WAIT_READ, COALESCE_WRITE, WAIT_WRITE, WAIT_SHARED, ATOMIC_*`).

- **Reads:** the lowest pending lane selects a 128-bit block; every pending lane whose word lies in that block is served by the same transaction. Lanes in other blocks need further transactions.
- **Writes:** all pending lanes that map to the selected block are packed into one 128-bit beat with a 4-bit word strobe, so contiguous lane addresses produce a single block write.
- **Atomics** (`ATOM_ADD`, `ATOM_CAS`) are serialised lane by lane: read the block, apply the operation, write back one word, return the old value in `rd`.
- The effective address is `rs + sign_extend(imm16)` for `LDR`/`STR`; atomics and shared-memory accesses use `rs` alone.

3.5.2 Banked shared memory

64 words in 4 banks of 16 words: `bank = addr[1:0]`, `offset = addr[5:2]`. Every lane has its own request port. Per cycle, lanes are processed in index order and a lane is deferred (not dropped) if

- *write bank conflict*: a lower lane writes the same bank in the same cycle;
- *read bank conflict*: a lower lane reads the same bank at a *different* address;
- *read-after-write*: a lower lane writes the address being read.

Lanes reading the *same* address as a lower lane are served by **multicast broadcast** (they copy the data fetched for that lower lane). Shared memory resets to zero. The memory is shared by all warps of the core; ordering between warps requires `SYNC`.

3.5.3 Caches and victim write buffer

The L2 must not push to a full buffer: a push is attempted only when the buffer reports space *and* no push is already in flight (this handshake was a bug, B4, Chapter 5); the buffer additionally asserts an error if a push ever arrives while full. At the end of a launch the D-cache is invalidated and the L2 flushed, after which external memory holds all results.

3.6 Exceptions

Division by zero (cause 1), out-of-range addresses (global $>$`0xFFFF`, shared $\geq$64; cause 2) and a `SYNC` executed while the warp's divergence stack is non-empty (cause 3) mark the offending warp `FAULTED` and flush it. The other warps continue and a faulted warp counts as finished for DONE. The first fault is reported to the host: STATUS[4] and `gpu_fault_pin` go high and FAULT_INFO holds its cause, warp and PC (§2.2.1); write 1 to STATUS[4] to clear. With several cores only the lowest-numbered faulting core is reported.

Level	Organisation	Policy
I-cache	16 lines, direct-mapped, 4-word lines	read-only; miss fetches the block from the program controller
D-cache (L1)	16 lines = 8 sets × 2 ways, LRU bit	**write-through**; write hit merges into the line, write miss does not allocate; read miss allocates
L2	64 lines = 16 sets × 4 ways, 3-bit PLRU, 4 sectors of 32 bit per line with per-sector dirty bits	**write-back, write-allocate**; evicting a dirty line pushes it into the victim write buffer
Victim write buffer	4 entries (128-bit data + 4 dirty bits)	merges pushes to the same address (dirty-sector merge); drains when half full, on a flush, or after an idle timeout (31 cycles); an L2 miss that hits the buffer is served from it and the entry is popped (probe-hit bypass)

Table 3.2: Cache hierarchy (ASIC profile).

3.7 Performance monitoring unit

Four 32-bit counters, each selecting one of 32 event lines (Appendix A) through DCR 0x10–0x13, with reset and snapshot pulses. L2 hit/miss events read zero (L4), and the PMU registers are not reachable from the Wishbone interface (L13).

Chapter 4

Instruction set architecture

4.1 Instruction formats

All instructions are 32 bits. The opcode occupies bits [31:26].

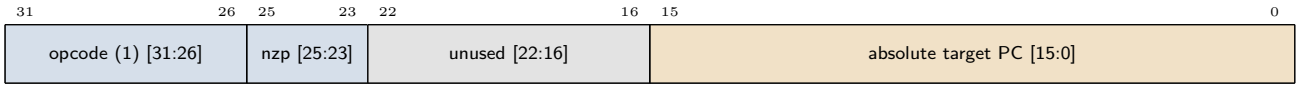
R-type (register–register)



I-type (immediate, memory)



Branch type (`BRnzp`, `CALL`)



`CALL` uses opcode 13 with bits [25:16] unused. `RET_FN`, `EXIT`, `SYNC` and `NOP` carry only the opcode.

Load-upper-immediate (`LUI`)



`rd = imm20 « 12`.

Operand-slot conventions

For `STR` and `STSH` the register holding the *value to store* is encoded in the `rd` field and the address register in `rs` (the decoder re-routes the `rd` slot into the second source). For `MAC`, `FMA` and `ATOM_CAS`, `rd` is also an input. `BRnzp` encodes the condition in `[25:23]` as `N=bit 25, Z=bit 24, P=bit 23`.

4.2 Register file conventions

Each lane holds 2 warps × 32 = 64 registers. The register file is written by up to three sources per cycle (Chapter 3); a write-through bypass forwards a value to a same-cycle read of the same warp.

4.3 Instruction reference

Notation: `rd, rs, rt` are registers, `imm` a 16-bit signed immediate, `T` an absolute program address. Integer values are 32-bit two's complement unless stated.

Register	Name	Access	Value / convention
R0–R28	general purpose	R/W	reset to 0; no ABI is imposed by the hardware
R29	`blockIdx`	R, writes ignored	block id assigned by the dispatcher (8 bits, zero-extended)
R30	`blockDim`	R, writes ignored	`THREADS_PER_BLOCK` × `NUM_WARPS` = **8** (threads per block)
R31	`threadIdx`	R, writes ignored	`warp_id` × 4 + `lane` (0–7)

Table 4.1: Per-thread register file. Global thread id: `MUL R0,R29,R30 ; ADD R0,R0,R31`.

4.3.1 Integer arithmetic and logic

Mnemonic	Op	Operands	Behaviour
ADD	3	rd, rs, rt	$rd = rs + rt$ (wraps)
SUB	4	rd, rs, rt	$rd = rs - rt$ (wraps)
MUL	5	rd, rs, rt	low 32 bits of the signed product
DIV	6	rd, rs, rt	unsigned quotient; `rt=0` faults. Multi-cycle: shared iterative divider, up to 33 cycles per active lane (R6)
MOD	22	rd, rs, rt	unsigned remainder; same caveats as `DIV`
AND	17	rd, rs, rt	bitwise AND
OR	18	rd, rs, rt	bitwise OR
XOR	19	rd, rs, rt	bitwise XOR
SHL	20	rd, rs, rt	logical left shift by `rt`; counts ≥ 32 give 0
SHR	21	rd, rs, rt	logical right shift; counts ≥ 32 give 0
MIN	23	rd, rs, rt	signed minimum
MAX	24	rd, rs, rt	signed maximum
ABS	25	rd, rs	signed absolute value (`0x80000000` $\mapsto$ `0x80000000`)
NEG	26	rd, rs	two's-complement negate
MAC	27	rd, rs, rt	$rd = rd + rs \cdot rt$ (signed, low 32 bits); reads `rd`
POPCNT	30	rd, rs	number of set bits
CLZ	31	rd, rs	leading zeros (CLZ(0) = 32)
BREV	32	rd, rs	bit reversal
CONST	9	rd, imm	$rd = \text{sext}(imm16)$
LUI	29	rd, imm20	$rd = imm20 \ll 12$

The ALU also computes carry, overflow, zero, negative and saturation flags, but **no instruction reads them**.

4.3.2 Floating point (single precision)

Mnemonic	Op	Operands	Behaviour
FADD	33	rd, rs, rt	$rd = rs + rt$
FSUB	34	rd, rs, rt	$rd = rs - rt$
FMUL	35	rd, rs, rt	$rd = rs \times rt$
FMA	36	rd, rs, rt	$rd = rs \times rt + rd$ (reads `rd`); single rounding point: truncation

4.3.3 Compare, branch and call

Mnemonic	Op	Operands	Behaviour
CMP	2	rs, rt	sets this lane's NZP flags: `N = rs<rt, Z = rs==rt, P = rs>rt` using an **unsigned** comparison (L5). Writes no register

Mnemonic	Op	Operands	Behaviour
BRnzp	1	nzp, T	taken iff (NZP & nzp) $\neq 0$. NZP is 0 after reset, so a branch before any CMP falls through even with nzp=111
CALL	13	T	push PC+1 on the lane's return stack, jump to T
RET_FN	14	—	pop the return stack and jump (undefined if empty)
EXIT	15	—	terminate the lane; if the warp's divergence stack is non-empty the next group resumes
NOP	0	—	no operation

4.3.4 Memory, shared memory and atomics

Mnemonic	Op	Operands	Behaviour
LDR	7	rd, [rs+imm]	load word from global memory; fault if address >0xFFFF; addresses ≥0x4000 alias (L14)
STR	8	rv, [rs+imm]	store rv (encoded in the rd slot) to global memory
LDSH	11	rd, [rs]	load from shared memory (no offset); fault if address ≥ 64
STSH	12	rv, [rs]	store rv (rd slot) to shared memory
ATOM_ADD	16	rd, rs, rt	old = mem[rs]; mem[rs] = old+rt; rd = old; lanes serialised
ATOM_CAS	28	rd, rs, rt	old = mem[rs]; if old = rd then mem[rs] = rt; rd = old (rd is both compare input and result)

4.3.5 Synchronisation

Mnemonic	Op	Operands	Behaviour
SYNC	10	—	block barrier (__syncthreads()): the warp waits until every non-finished warp of the core has executed SYNC. **Requires warp-uniform control flow** (§4.5)

Opcodes 37–63 are reserved and decode as inert instructions.

4.4 Programming example

The following program is the instruction sequence of a branch/call test; the encodings were generated by the project's assembler functions (scripts/isa.py) and executed on the RTL.

```
0:  24200007  CONST R1, 7
1:  145f0800  MUL   R2, R31, R1      ; R2 = threadIdx * 7
2:  08020800  CMP   R2, R1           ; N/Z/P = R2 <,==,> R1 (unsigned)
3:  04800005  BRp   5                ; skip the call when R2 > R1
4:  34000007  CALL  7
5:  205f012c  STR   R2, [R31+300]    ; mem[300+tid] = R2
6:  3c000000  EXIT
7:  0c420800  ADD   R2, R2, R1       ; callee
8:  38000000  RET_FN
```

4.5 Architectural restrictions and hazards

R1 — `SYNC` requires warp-uniform control flow

The barrier is tracked at **warp** granularity: when a lane group executes `SYNC`, the warp enters `WAITING_BARRIER` and is counted once (`barrier_count >= num_active_warps_comb`, `scheduler.sv:279`). Because divergent lane groups are serialised until `EXIT` (no join-point reconvergence), a `SYNC` that follows divergent code is executed once *per group*, and the first group to arrive releases the barrier while other lanes have not yet performed their stores. This used to let threads read stale shared memory (observed: 4 of 8 threads in the directed test, and in 18 of 240 early random programs); the hardware now detects it and faults the warp with cause 3 instead (the kernel is rejected, not silently wrong). **Rule:** every branch and loop bound that precedes a `SYNC` must be uniform across the warp (derived from constants, `R29`, `R30`, or the warp id); it may differ *between* warps. Divergence without barriers is fully supported. A `SYNC` executed after some lanes have already executed `EXIT` is likewise unsupported.

R2 — return-stack overflow

Nesting deeper than 8 `CALL`s wraps the 3-bit stack pointer silently and corrupts the oldest return address; a ninth nested call hangs or mis-returns on the RTL. `RET_FN` with an empty stack jumps to a stale address. Keep call depth ≤ 8.

Other rules

R3 `CMP` is unsigned. **R4** A conditional/unconditional `BRnzp` needs a preceding `CMP` (NZP resets to 0); an "always" jump is `CMP` of a register with itself followed by `BR 111`. **R5** Do not write `R29`–`R31`. **R6** `DIV`/`MOD` are slow (shared iterative divider: up to 33 cycles per active lane, the warp waits) and need a non-zero divisor. **R7** Division by zero, bad addresses and `SYNC` while diverged stop the warp; check STATUS[4] after DONE. **R8** Results are guaranteed in external memory only after DONE. **R9** Do not rely on the old value returned by `ATOM_ADD` being a particular order across lanes.

Chapter 5

Verification and physical implementation

5.1 Verification methodology

1. **Lint:** Verilator `-Wall` on the sim, ASIC and FPGA profiles; ten baseline warning classes are waived, any new class fails.
2. **Unit tests** (Verilator, self-checking): ALU (56 736 vectors, run with and without `USE_COMB_DIV`), decoder (all 64 opcodes), `pc`, register file, FMA (4 013 pipelined results against a double-precision reference), D-cache and L2+VWB (random partial-write traffic with random memory latency against a golden memory).
3. **Directed system tests:** NZP/branch/nested-call test on the ASIC profile; 2-core, 32-thread 5×5 matrix multiply (CALL/RET, divergence, DIV).
4. **SVA checkers** bound into the RTL (no source edits): issued PC equals fetched PC, no dropped instruction, return-stack pointer discipline, CMP updates NZP, divergence-stack bound, LSU strobe sanity, fetcher overlap, VWB no-push-while-full.
5. **Golden reference model and constrained-random fuzzing.** A Python interpreter (`Machine` in `isa.py`) executes the same machine words, including a bit-exact port of the FMA datapath. The generator (`gen.py`) emits terminating, race-free programs using structured control flow (forward if/else, loops with per-thread or per-warp trip counts, an acyclic call graph), integer and FP arithmetic, global and shared memory, `SYNC` barriers and `ATOM_ADD`; each program dumps all registers to memory. The runner (`rand_regress.py`) executes each program on the RTL with randomised memory latency and compares *all* 1024 data words with the model. Programs that use barriers are generated in a "uniform" mode (restriction R1) with deliberate warp imbalance.
6. **Limitation characterisation** (`make limits`) and a **documentation consistency check** (`make docs-check`) keep this manual and the repository synchronised.

`make verify` runs lint, unit, system, a short random run, limitation and documentation checks.

5.2 Regression results

Campaign (generator content)	Memory latency (max extra cycles)	Programs / latency	Mismatches
Integer, control flow, CALL/RET	0, 2, 5, 10, 25	600 each (3000)	0
+ FP, shared memory, SYNC, atomics	0, 5, 20	1200 each (3600)	0
+ warp-imbalanced barriers	0, 3, 10, 30	800 each (3200)	0
Same generator, **current RTL** (L2 data in SRAM)	0, 3, 10, 30	750 each (3000)	0
Same generator, current RTL, other seeds	0, 2, 5, 10, 25, 40	1700 each (10200)	0

Table 5.1: Constrained-random campaigns (ASIC profile, 8 threads). The first three ran before the L2 data array moved into an SRAM macro; the last two ran on the current RTL. Each program is 150–300 instructions.

All 22 800 runs matched the model bit-for-bit with all SVA checkers enabled. The first campaign pre-dates the FP/shared-memory/atomic extension of the generator; the latency sweep requested for sign-off (0, 2, 5, 10, 25) was therefore also covered by the later campaigns, and the last campaign repeats it on the current RTL. At chip level (`make top-quick`) 40 seeds, two kernels each with a soft reset between them, ran purely through the Wishbone pins and matched the model; the same test passes against the real OpenRAM model (`make top-macro`), and a zero-delay gate-level simulation of the synthesised netlist matched for 200 seeds (`make gls`; pre-placement, no SDF, two-state). Mutation scores were not re-measured after the L2 change.

5.3 Mutation testing

Faults were injected into copies of the RTL and the random regression was run on 100 programs at 4 cycles of latency (200 for the barrier fault). The figure is the number of programs whose result differed from the model. All faults were detected except those marked. Additional ALU, decoder, register-file and FMA mutants were killed

Injected fault	Detected by	Comment
`FSUB` sign not inverted	18 / 100	also unit test
FP normalisation exponent off by one	54 / 100	also unit test
`ATOM_ADD` adds a constant instead of the operand	55 / 100	
Shared-memory store uses lane 0's data for all lanes	21 / 100	
Barrier released when the first warp arrives	18 / 200	**0 / 100** before warp-imbalanced workloads were added
Register-file ALU-vs-FP collision guard removed	0 / 100	survived; design argument in Chapter 3
FMA alignment threshold 72→40	not detected	effect lies below the 24-bit truncation (unobservable)
Original bugs re-introduced (B1–B4): dead NZP/CALL/RET; ID-latch drop; fetcher/warp skew; L2 push-while-full	98 / 92 / 98 / 100 of 100	each also caught by a unit or system test and by a specific SVA

Table 5.2: Mutation-testing scorecard.

by their unit tests (e.g. overflow flag, CLZ direction, signed MIN, STR operand slot, immediate sign-extension, FP priority, r30/r31 protection).

5.4 Bugs found and fixed during verification

ID	Fix commit	Summary
B1	`499a7d0`	ASIC port deleted the NZP, return-address and stack-pointer updates in `pc.sv`; branches and calls were dead.
B2	`0ab48c4`	ID-latch squash for the previous occupant's warp discarded a fresh instruction of another warp; the fetcher had already consumed it.
B3	`ab57e33`	Fetcher held an instruction for one warp's PC while the scheduler issued another warp: wrong instruction issued; fixed with a PC tag and refetch.
B4	`e086fc4`	L2 pushed to a full victim write buffer (stale ready); the dirty line was dropped.

Table 5.3: Fixed bugs. B2–B4 are latent in the shared design and are fixed in the ASIC copy only (erratum L10).

5.5 Physical implementation

Status — the mini profile is routed with per-corner SRAM timing; the full profile is not

The **mini profile** (`MINI=1`: 2 lanes × 2 warps, 32 words of shared memory) has a routed layout at the 40 ns clock (`flow/runs/place_1029_1604`) with: routing DRC 0, antenna 0, LVS 0 (macros compared at their pins), KLayout DRC 0, and a full-chip Magic DRC with 0 integration errors (the OpenRAM macros' own 12 984 239 violation boxes are waived, see below). Setup is met with ≥ +3.27 ns and hold with ≥ +0.27 ns in all nine corners, using SRAM liberty files characterised with OpenRAM at the TT, SS and FF corners. The Magic-versus-KLayout XOR shows 20 differences, all on the cell-boundary marker layer 235/4 (not a mask layer). **Not done:** the full-size profile has never been routed (it needs a rectangular die inside the 2.92 mm × 3.52 mm Caravel user area and probably more than 16 GB of memory); the macros are black boxes in LVS; the SRAM read-port (`dout1`) nets exceed the macro's own 27.6 fF capacitance limit (36–39 pins; slow-corner slew up to 6 ns, extrapolated; waiver recorded; erratum L17), and 642 capacitance and 524 fanout violations remain; there is no gate-level simulation of this netlist, no equivalence check, and no Caravel wrapper.

5.5.1 Flow and configuration

`run_asic.sh` converts the SystemVerilog to Verilog with `sv2v` (`-DSYNTHESIS -DTARGET_ASIC=1`, plus `-DTARGET_ASIC_MINI=1` for the mini profile), runs a Yosys elaboration/latch audit (zero latches required), then runs LibreLane 3.0.16 (OpenROAD-based) in a container with `flow/config.yaml` (`MINI=1 ./run_asic.sh` reproduces the reference result: 85 of 85 compared metrics identical to the hand-assembled first run):

Parameter	Value
Design / top	`gpu_asic_top`
Clock	port `clk`, 40.0 ns, 0.25 ns uncertainty, 5 % OCV derate; clock source modelled as `clkbuf_16`
Die / core area (mini)	[0,0,2400,2400] µm / [50,50,2350,2350] µm, absolute sizing; placement density target 30 %
PDK / library	`sky130A` (version pinned in the Makefile) / `sky130_fd_sc_hd`
Synthesis	`SYNTH_STRATEGY "DELAY 4"` (the only strategy whose netlist met 40 ns before placement; +7.8 % area)
Macros	three `sky130_sram_2kbyte_1rw1r_32x512_8` (PMEM window, DMEM window, L2 data array), per-corner liberty (TT, SS, FF, characterised with OpenRAM); power connections `VPWR VGND vccd1 vssd1`; 12 µm metal-4 keep-out bands at the macro pin edges while the power grid is generated
Repair and timing	post-routing slew/cap repair (margins 70 %/70 %, 1 200 µm wire-length cap); post-CTS hold margin 0.4 ns; timing repair again after the slew repair (hold margin 0.4 ns); antenna repair up to 8 rounds at 25 %
Stream-out and DRC	KLayout GDS is the deliverable (one top cell, `KLAYOUT_CONFLICT_RESOLUTION OverwriteCell`); flow Magic DRC non-fatal, verdict by `make signoff-drc`

5.5.2 Results of the final mini layout (`flow/runs/place_1029_1604`)

The OpenRAM macros are not clean under Magic's sky130 rules (5 579 171 boxes in 31 rule types on one macro alone). A box is waived only if its rule also fires on the standalone macro and it lies inside a macro outline; whether a shuttle accepts these macros is the operator's decision (erratum L17 and the project's open-problems list). KLayout's deck reports no violation on the macro.

Metric	Value
Instances	987 884 in total: 205 333 standard cells (1.61 mm^2), 62 490 tap cells, 782 548 fill cells, 3 macros (0.85 mm^2); of the standard cells 14 732 flip-flops, 7 187 antenna diodes, 3 069 clock buffers
Die / utilisation	2.4 mm × 2.4 mm; standard-cell utilisation 36 %
Routing	9.36 m of wire, 1 050 777 vias, 0 DRC errors, 0 antenna violations
LVS / XOR / KLayout DRC	0 / 20 (layer 235/4 boundary marker only) / 0
Magic DRC, full chip	12 984 239 boxes, all inside the three macros and in rules that also fire on the standalone OpenRAM macro; **0 integration errors** (`make signoff-drc`)
Slew, slow corner	6 300 pins above the flow's 0.75 ns target, 278 above the library's 1.5 ns range, 128 above 2 ns (worst 6.0 ns, on SRAM `dout1` outputs at loads beyond the library table); clock-root input about 1 ns
Capacitance / fanout violations	642 / 524 (open); 36–39 are SRAM `dout1` pins above the macro's 27.6 fF limit (waived)
IR drop (worst / average)	0.9 mV / 19 µV
Power (vectorless estimate)	39.9 mW

5.5.3 Static timing across PVT corners

Worst slack from OpenSTA on the routed design (clock 40 ns, 0.25 ns uncertainty, 5 % derate). Corner names: `max/nom/min` = parasitic extraction corner, `ss/tt/ff` = process/voltage/temperature.

Corner	Process	V / T	Setup WS (ns)	Hold WS (ns)
`max_ss_100C_1v60`	slow	1.60 V, 100 °C	**3.274** (worst setup)	1.462
`nom_ss_100C_1v60`	slow	1.60 V, 100 °C	3.932	1.486
`min_ss_100C_1v60`	slow	1.60 V, 100 °C	4.554	1.509
`max_tt_025C_1v80`	typical	1.80 V, 25 °C	13.624	0.565
`nom_tt_025C_1v80`	typical	1.80 V, 25 °C	13.693	0.593
`min_tt_025C_1v80`	typical	1.80 V, 25 °C	13.829	0.604
`max_ff_n40C_1v95`	fast	1.95 V, −40 °C	15.753	**0.271** (worst hold)
`nom_ff_n40C_1v95`	fast	1.95 V, −40 °C	15.830	0.297
`min_ff_n40C_1v95`	fast	1.95 V, −40 °C	15.940	0.309

Table 5.4: Slack by corner of the final mini layout. The SRAM macros are timed with liberty files characterised per corner (TT, SS, FF) with OpenRAM; the earlier typical-only library was 4.7 times too fast (erratum L17).

5.5.4 Outstanding sign-off items

A transistor-level LVS of the macro; the shuttle's decision on the macro DRC waiver; the SRAM `dout1` output nets above the macro's capacitance limit, and the other capacitance and fanout violations; gate-level simulation of the final netlist (zero-delay, then with the SDF files the flow writes), RTL-to-netlist equivalence; a routed full-size profile; the Caravel wrapper, harness simulation and precheck.

Chapter 6

Errata and known limitations

Each entry has a characterisation test or static check in the repository (`make limits`, `make docs-check`); a change in behaviour fails those checks so this chapter is updated. Severity: **H** may give silently wrong results or blocks use, **M** surprising but contained, **L** inconvenience.

ID	Sev	Title	Description / workaround
L1	M	No join-point reconvergence	Lane groups run the rest of the program serially; a `SYNC` after divergence faults (cause 3) instead of releasing early. Use warp-uniform control flow before every barrier (R1).
L2	–	`DIV`/`MOD` stubbed	**Resolved:** computed by a shared iterative divider (R6).
L3	–	Memory at the chip top (resolved)	Previously tied off. The window at `0x3000_1000` is implemented on two OpenRAM macros; see L15–L17 for what remains.
L4	L	L2 PMU events zero	L2 hit/miss/evict event outputs are constants.
L5	M	`CMP` unsigned; NZP resets to 0	See rules R3, R4.
L6	M	8-deep call stack wraps	Silent corruption on the ninth nested call.
L7	L	4-deep divergence stack	Cannot overflow with four lanes; could with ≥ 6 lanes.
L8	M	FP is not IEEE-754	Truncation, flush-to-zero, no NaN/Inf, zero-operand alignment quirk.
L9	M	Atomics coverage	`ATOM_CAS`, ordered old values and multi-core coherence untested by random tests.
L10	M	Fixes only in ASIC copy	B2–B4 are not applied to the generic `src/rtl`.
L11	L	Test debt	Legacy testbenches do not build against the current block interface; `tb_matmul` times out under iverilog (passes under Verilator).
L12	–	Faults invisible	**Resolved:** STATUS[4], FAULT_INFO and `gpu_fault_pin` report the first fault. The faulted warp's later results are still missing; check STATUS[4] after DONE.
L13	L	PMU/DCR unreachable	Only DCR address 0 is driven by the Wishbone wrapper.
L14	M	Address aliasing	The fault limit (`0xFFFF`) exceeds the 14-bit bus: accesses in `0x4000`–`0xFFFF` wrap onto `addr & 0x3FFF` without a fault. Keep global addresses below `0x4000`.
L15	M	Only 512+512 words on chip	GPU program and data words $\geq$512 read as 0 (instruction fetch) or read 0 / drop stores and set STATUS[2] (data). Programs must fit 512 words; the random generator uses a 32-word private region per thread.
L16	M	Host access exclusive; soft reset to re-run	Window accesses are refused while BUSY. Caches are flushed but not invalidated after a run, so write CTRL[1] (soft reset) before loading a new kernel.
L17	M	SRAM read-port outputs exceed the macro's capacitance limit	Per-corner liberty files now exist (the shipped typical-only file was 4.7 times too fast, clock to output 0.38 ns against 1.81 ns), and timing is met in all nine corners. After routing, 36–39 `dout1` pins carry 40–70 fF against the macro's 27.6 fF limit; the slow-corner slew there reaches about 6 ns (extrapolated). Waiver recorded, not fixed.

Pin count: the wrapper has 108 signal pins; meeting a smaller GPIO budget (for example a 38-pin user area) would require a narrower address/data interface.

Appendix A

PMU event map

The 32-bit event bus (`core.sv`, `event_bus`) feeds four counters; each counter selects one bit with a 5-bit selector.

Bit	Event	Bit	Event
3	cycle (core running)	16	scheduler stall: barrier
4	active (EX has active lanes)	17	scheduler stall: no ready warp
5	instruction issued	18	D-cache read access
6	fetch stall	19	D-cache read hit
7	flush	20	D-cache read stall
8	memory operation started	21	D-cache write access
9	I-cache access	22	D-cache write hit
10	I-cache hit	23	D-cache write stall
11	I-cache stall	24	FP instruction fired
12	scheduler idle	25	FP write-back valid
13	warp switch	26	FP RAW stall
14	divergence	27	FP write-back collision
15	scheduler stall: memory	0–2, 28–31	unused (0)

Appendix B

Provenance and how to re-check this document

Source: repository commit `dadec7f` (RTL under `ASIC/src/rtl`, flow under `ASIC/flow`); physical numbers from `ASIC/flow/runs/RUN_2026-10-19_14-54-17`. The repository documentation set in `ASIC/docs` (pages 01–08) contains line-level citations for every statement here. To re-audit: run `make verify` in `ASIC/`; `make docs-check` validates generated tables and citations; `make limits` validates Chapter 6; `make datasheet` rebuilds this PDF.